Architecture-Agnostic Self-Supervised Diffusion MRI Denoising with Hybrid Random Gradient Subsets

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Abstract

Diffusion-weighted magnetic resonance imaging (DW-MRI) provides clinically and scientifically relevant measurements of brain tissue microstructure, but its reliability is limited by low signal-to-noise ratio, acquisition-dependent noise, and the high cost of repeated measurements. Self-supervised denoising is attractive in this setting because clean reference images are rarely available. This manuscript evaluates Hybrid Random Gradient Subset (Hybrid RGS), an architecture-agnostic self-supervised DW-MRI denoising framework that combines non-target angular context across diffusion gradients with voxel-wise masking within the target volume. The framework is instantiated with convolutional DRCNet, Restormer-style, and lightweight two-dimensional neural backbones to test whether the same sampling-and-masking objective transfers across model families rather than to claim superiority for a single architecture. On D-Brain with synthetic Rician noise, Hybrid RGS improves substantially over Patch2Self and MD-S2S in image-domain ROI metrics, while MP-PCA remains a strong baseline for FA preservation under the controlled synthetic noise model. Ablations show that random gradient sampling improves over deterministic sequential windows and that $K = 16$ is used as a fixed cross-model operating point that is favorable to DRCNet; Restormer's strongest result in the K sweep occurs at $K = 10$ (PSNR-ROI = 26.60 dB), so the fixed $K = 16$ row is not Restormer-optimal and the K recommendation should not be treated as uniformly optimal across backbones. FiLM conditioning is evaluated as a lightweight metadata-conditioning extension; it improves full-volume PSNR but slightly degrades ROI PSNR in the current

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registry, so it is treated as exploratory rather than uniformly beneficial. A controlled capacity-and-dimensionality study at the fixed $K = 16$ operating point indicates that lightweight 2D baselines (Plain-CNN-2D, Res-CNN-2D) can outperform the current volumetric 3D variants on most scalar image-domain metrics and on DRCNet tensor metrics, while Restormer3D retains a slight FA-MAE advantage; this comparison holds at $K = 16$ and should not be extrapolated to Restormer's best-sweep configuration at $K = 10$, supporting the interpretation that the Hybrid RGS objective is the dominant contributor in these experiments. Evaluation on Stanford HARDI demonstrates feasibility on real scanner data without ground truth and shows that the framework scales to larger gradient shells.

Keywords: Diffusion-weighted MRI, DW-MRI, Self-supervised denoising, J-invariance, Random gradient subset, Architecture-agnostic learning, Blind-spot denoising

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1. Introduction

Diffusion-weighted MRI is widely used to characterize brain tissue microstructure through the measurement of water diffusion [3]. By acquiring multiple diffusion-weighted volumes across gradient directions and b-values, DW-MRI enables downstream estimation of apparent diffusion coefficient (ADC), fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD), radial diffusivity (RD), and tractography-derived biomarkers. Despite this importance, DW-MRI is particularly vulnerable to low signal-to-noise ratio, Rician or noncentral-chi noise, eddy-current artifacts, susceptibility distortions, and patient motion. Noise reduction is therefore a critical preprocessing step, especially for high angular resolution protocols and quantitative studies.

Classical denoising methods, such as local low-rank and MP-PCA approaches, exploit redundancy in spatial neighborhoods and diffusion volumes. More recent self-supervised methods exploit the fact that clean ground truth is often unavailable but repeated structure exists across voxels, slices, and diffusion gradients. Patch2Self introduced a volume-wise J-invariant strategy in which each diffusion-weighted volume is predicted from the remaining volumes. Self2Self-style methods instead mask pixels or voxels and train a network to infer the hidden values from their context. These ideas are complementary: DW-MRI contains both angular redundancy across diffusion gradients and spatial redundancy within each 3D brain volume.

This work studies a Hybrid Random Gradient Subset (Hybrid RGS) framework for self-supervised DW-MRI denoising. Rather than predicting each target volume from all remaining gradients in a deterministic leave-one-out manner, Hybrid RGS samples random subsets of diffusion gradients during training and inference. Within each sampled subset, the target volume is partially masked so that the model must predict masked voxels using both neighboring unmasked spatial information and auxiliary diffusion volumes. This design provides a practical compromise between strict J-invariance, memory-efficient training, and architecture-agnostic deployment across compact convolutional, transformer-style, two-dimensional, and three-dimensional denoisers.

The main contributions of this manuscript are:

- an architecture-agnostic Hybrid RGS framework that combines volume-wise gradient exclusion with voxel-wise masking for self-supervised DW-MRI denoising;
- implementations with DRCNet, Restormer-style, and two-dimensional backbones to test whether the same sampling-and-masking objective supports multiple model families;
- a controlled comparison against MP-PCA, Patch2Self, MD-S2S, two-dimensional neural variants, and supervised reference models;
- an evaluation strategy emphasizing both image-domain metrics and diffusion-derived biomarkers such as FA, MD, AD, and RD;
- ablation studies quantifying random versus sequential sampling, input subset size K, gradient-direction conditioning, noise robustness, real-scanner generalization, and backbone sensitivity including 2D versus 3D processing.

2. Related Work

2.1. DW-MRI Denoising

DW-MRI denoising methods must suppress acquisition noise while preserving subtle contrast variations across diffusion directions. MP-PCA denoising estimates local signal subspaces and removes components attributed to random noise, making it a widely used preprocessing baseline for diffusion MRI [8]. Other classical methods exploit non-local similarity, local low-rank structure, or model-based priors. Although these approaches are effective and computationally practical, their assumptions may be challenged by spatially varying noise, protocol changes, and complex anatomical or pathological variability.

2.2. J-Invariant Self-Supervision

J-invariance provides a theoretical framework for denoising when clean targets are unavailable [10]. A function is J-invariant when its prediction on a subset of dimensions J does not depend on the observed noisy values in those same dimensions. If the noise is independent across the dimensions in the partition and the signal is statistically dependent, minimizing prediction error against noisy observations can calibrate a denoiser without clean references. In image restoration, the J-set can correspond to masked pixels or voxels. In DW-MRI, the J-set can also correspond to complete diffusion-weighted volumes, as in leave-one-gradient-out regression.

2.3. Patch2Self and Self2Self

Patch2Self applies volume-wise J-invariance to diffusion MRI by predicting each diffusion-weighted volume from the remaining volumes [9]. Its original formulation uses linear regression, although alternative regressors can be used. Self2Self trains neural networks from a single noisy image using Bernoulli masking and dropout-like stochasticity [13]. Hybrid RGS combines these two perspectives: it excludes information from selected target voxels while also exploiting the angular context of other diffusion volumes.

2.4. Neural Backbones for Restoration

Three-dimensional neural networks are natural for volumetric MRI because they preserve anatomical context across slices, but they are more expensive than two-dimensional slice-wise networks, especially for 4D DW-MRI data with many diffusion directions. Restormer-style architectures use efficient attention and gated feed-forward blocks for high-resolution image restoration [1], while compact convolutional networks can offer favorable speed-memory tradeoffs. This paper studies these alternatives within the same self-supervised Hybrid RGS framework, using architecture as an experimental factor rather than as the central contribution.

3. Materials and Methods

3.1. Data Representation

A DW-MRI acquisition is represented as a 4D array $Y \in \mathbb{R}^{N_x \times N_y \times N_z \times G}$, where G is the number of diffusion-weighted volumes or gradients and N_x, N_y, N_z are spatial dimensions. Let $y_g \in \mathbb{R}^{N_x \times N_y \times N_z}$ denote the noisy 3D volume associated with gradient g. When synthetic noise experiments are used, a clean or higher-quality reference x_g may be available for evaluation only. The proposed self-supervised training does not require x_g.

The current experiments include D-Brain data at $b = 2500$ s/mm^2 and Stanford HARDI data at $b = 2000$ s/mm^2, with additional noise-level studies under synthetic Rician corruption. The reported self-supervised objective does not use clean targets during training, but quantitative D-Brain evaluation uses the available reference after denoising.

The controlled synthetic experiments use one D-Brain digital phantom, an anatomically accurate brain simulation generated under a 3T scanner model [4]. The phantom was downloaded from Figshare, cropped to $128 \times 128 \times 96$ voxels, and used at $1.4 \times 1.4 \times 1.4$ mm^3 resolution with six $b = 0$ s/mm^2 volumes and $G = 60$ diffusion-weighted volumes at $b = 2500$ s/mm^2, with directions uniformly distributed on the unit sphere. The real-data experiment uses one in vivo Stanford HARDI acquisition from a GE Discovery MR750 3T scanner, obtained through `dipy.data.fetch_stanford_hardi`, with ten $b = 0$ s/mm^2 volumes and $G = 150$ uniformly distributed diffusion-weighted directions at $b = 2000$ s/mm^2 in native 2 mm isotropic voxels [5]. For each dataset, each diffusion-weighted volume is independently min–max normalized to $[0, 1]$ as $Y'_g = (Y_g - \min(Y_g))/(\max(Y_g) - \min(Y_g) + \epsilon)$ with $\epsilon = 10^{-6}$, where the min and max are computed over all voxels in volume g. Synthetic Rician noise is added only to normalized D-Brain volumes. D-Brain training uses progressive 3D patches: 32^3 with stride 8, 24^3 with stride 6, and 16^3 with stride 4. Patches satisfying $\max(x_{\text{patch}}) \leq 0.02$ are discarded. No conventional train/validation/test split is used because training is self-supervised on the single subject and evaluation uses the clean phantom; Stanford HARDI is used only for feasibility without ground truth. The ROI brain mask contains voxels where any diffusion volume exceeds 0.02 and is used for PSNR-ROI, SSIM-ROI, and tensor-metric errors.

3.2. Hybrid RGS Formulation

Let t denote the physical target diffusion volume. Throughout the manuscript, K denotes the total number of input channels. For Hybrid RGS, one channel contains the masked target volume and the remaining $K-1$ channels contain non-target context gradients. Hybrid RGS samples a context set $C_t \subset \{1,\ldots,G\}\setminus\{t\}$ containing $K-1$ auxiliary gradients. Let $C_t = (c_1,\ldots,c_{K-1})$ denote an ordered tuple of these context indices (ordering is implementation-defined; channels are shuffled per training step). The network input is formed by concatenating the context volumes followed by the masked target volume:

$$z_t = \mathrm{concat}(y_{c_1},\ldots,y_{c_{K-1}},\tilde{y}_t), \tag{1}$$

where $\tilde{y}_t$ is a masked version of the target volume. Channels are zero-indexed in the implementation: after $K-1$ context channels (indices $0,\ldots,K-2$), the target channel is placed at index $K-1$, which corresponds to the Kth channel in one-based mathematical notation. A Bernoulli mask $m \in \{0,1\}^{N_x\times N_y\times N_z}$ is sampled with masking probability p, and the target channel is corrupted by replacing masked voxels with zeros:

$$\tilde{y}_t = (1-m)\odot y_t + m\odot r. \tag{2}$$

Here r denotes the replacement signal and $\odot$ denotes element-wise multiplication.

In all reported experiments, Bernoulli masks are sampled independently per voxel with probability $p = 0.3$, so $\Pr(m_i = 1) = 0.3$ for occluded supervised sites and $\Pr(m_i = 0) = 0.7$ for visible sites. The replacement signal is fixed to zero, $r = 0$, giving the implemented target-channel input $\tilde{y}_t = (1-m)\odot y_t$; zero replacement is used throughout training and inference, with no random or interpolation-based replacement. The mask m and the context indices C_t are not passed as explicit arguments to the network; the network receives only the K-channel tensor z_t, with masked voxels appearing as zeros and context identities implicit in the channel positions. The mask is applied only to the target channel at index $K-1$, while all $K-1$ context channels remain unmasked. Training uses the progressive patch schedule 32^3, 24^3, and 16^3 voxels, and the penalty is squared error, $\rho(u) = u^2$, with no robust loss.

The model f_θ predicts a denoised estimate of the target volume from the K-channel input stack only—the mask m and context indices C_t are not explicit network arguments. For the unconditioned baseline the prediction is

$\hat{y}_t = f_\theta(z_t)$. When FiLM gradient-direction conditioning is used (Section 4.9), the target gradient's acquisition metadata $q_t = [\cos_x, \cos_y, \cos_z, b_{\text{norm}}]$ is additionally provided to the FiLM layers:

$$\hat{y}_t = f_\theta(z_t, q_t). \tag{3}$$

The self-supervised loss is computed only on masked voxels:

$$\mathcal{L}_{\text{RGS}}(\theta) = \frac{1}{|\Omega_m|} \sum_{i \in \Omega_m} \rho\left(\hat{y}_{t,i} - y_{t,i}\right), \tag{4}$$

where $\Omega_m = \{i : m_i = 1\}$ and $\rho(u) = u^2$ in the reported experiments. This objective prevents the network from directly copying the observed noisy values at supervised voxels while preserving access to unmasked spatial context and randomly sampled angular context.

3.3. Connection to J-Invariance

Hybrid RGS uses two coupled self-supervised mechanisms. First, at the angular level, the prediction of target volume t is conditioned on a subset of non-target diffusion volumes C_t, following the principle that a target diffusion image should be inferred from other correlated measurements. Second, at the spatial level, the supervised voxels of the target channel are masked, so the training loss at those voxels cannot directly depend on their observed noisy values. The framework therefore interpolates between Patch2Self-like volume exclusion and Self2Self-like voxel masking. The masked training loss enforces a blind-spot objective at supervised masked sites: strictly, J-invariance holds for the training loss at the masked locations Ω_m. The final inference output, however, is a Monte Carlo masked ensemble (Eq. 5) and should not be described as strictly voxel-wise J-invariant at every output site—at inference each voxel is unmasked in many ensemble members, so predictions at those sites can depend on the observed noisy target value in those members.

The size and structure of the J-set induce a bias-variance tradeoff. If too much target information is hidden, the estimator may become biased because relevant spatial context is removed. If too little information is hidden, self-supervision becomes weak and the model may learn identity-like behavior. Hybrid RGS exposes this tradeoff through the subset size K, the masking probability p, the number of sampled contexts during inference, and the number of stochastic predictions averaged per context.

3.4. Architecture-Agnostic Instantiations

3.4.1. DRCNet-hybrid-RGS

The first instantiation adapts the Denoising Recurrent Convolutional Network (DRCNet) architecture [2], originally proposed for brain MRI Rician denoising, for volumetric DWI restoration under the Hybrid RGS framework. Architecturally, DRCNet uses a single-level encoder-decoder (one downsampling and one upsampling stage) rather than a multi-level hierarchy, with channel width held constant at `base_filters` throughout; this contrasts with the three-level, channel-doubling hierarchy of Restormer-hybrid-RGS described below. The core processing stage is a Recurrent Convolutional Denoising Block (RCDB): a gated-recurrent-unit-style module that iteratively refines a downsampled feature map using dense asymmetric factorized convolutions (parallel $3 \times 1 \times 1$, $1 \times 3 \times 1$, $1 \times 1 \times 3$ kernels fused by a $1 \times 1 \times 1$ convolution), reducing computation by approximately 60% relative to full $3 \times 3 \times 3$ convolutions. The original ablation study identified four unfolded RCDB iterations as the best accuracy-cost trade-off; this implementation preserves that choice, applying the gated block recurrently four times per forward pass.

Three modifications distinguish DRCNet-hybrid-RGS from the original architecture. First, and most significant, is the removal of Global Residual Learning (GRL). The original DRCNet formulates denoising as $G(x) = F(x)+x$: the network predicts the additive noise residual and adds it back to the noisy input, which requires matching input and output channel counts. Hybrid RGS instead uses `in_channels` $= K$ (context gradients plus the Bernoulli-masked target at index $K-1$) and `out_channels` $= 1$ (a single denoised target volume), so a direct channel-wise residual is undefined. The network therefore predicts the denoised target volume directly rather than a noise residual – a conceptually different learning target from the original formulation. Second, the base feature width is reduced from 64 feature maps (fixed in the original) to 32 (`base_filters=32`), trading capacity for memory given the additional context channels processed per 3D patch. Third, because the output no longer represents a signed residual, the final activation is configurable between `PReLU` (unbounded, matching the original) and `Sigmoid` (bounded to $[0, 1]$, matching the min–max normalized target range and reducing background bias).

As with Restormer-hybrid-RGS, an optional FiLM conditioning mechanism [12] is evaluated as an extension (Table 5): two FiLM layers modulate features at the bottleneck (after downsampling, before the RCDB recurrence) and at the decoder output (after upsampling, before the final concatenation), conditioned on the target gradient's $[\cos_x, \cos_y, \cos_z, b_{\text{norm}}]$ vector, adding approximately 3.9% parameters. All other components – the encoder (Conv3d 3×3×3 feature extraction followed by strided Conv3d 2×2×2 downsampling), the RCDB update/reset gate equations, and the decoder (ConvTranspose3d $2 \times 2 \times 2$ upsampling, skip concatenation, and $1 \times 1 \times 1/3 \times 3 \times 3$ output convolutions) are structurally unchanged from [2]. The resulting model contains approximately 0.116M parameters at $K = 16$ (Table 2). Figure 1 illustrates the DRCNet3D architecture and its lightweight 2D baseline counterpart.

3.4.2. Restormer-hybrid-RGS

The second instantiation adapts the 2D Restormer architecture [1] for 3D volumetric DWI denoising under the Hybrid RGS framework. Restormer's efficient attention mechanism and gated feed-forward network have demonstrated strong performance on high-resolution 2D image restoration tasks. The 3D adaptation balances representational capacity with memory constraints for volumetric transformer attention, while modifying the input/output interface to accommodate the K-channel random gradient subset sampling and blind-spot target masking objective.

The adaptation involves several architectural modifications. First, all 2D spatial operations are converted to their 3D equivalents: `Conv2d`→`Conv3d`, layer normalization reshaping from $(B, C, H, W) \to (B, HW, C)$ to $(B, C, D, H, W) \to (B, DHW, C)$, and attention spatial flattening over three dimensions. The Multi-DConv Head Transposed Self-Attention (MDTA) and Gated-DConv Feed-Forward Network (GDFN) blocks remain structurally identical to the original, with efficient transposed channel-wise attention and gated pointwise convolutions, but now operate on volumetric features.

Second, the hierarchy depth is reduced from the original four-level encoder-decoder (with three downsampling operations and bottleneck spatial compression of 512×) to a three-level architecture with two downsampling operations and 64× compression. The specific configuration uses `dim` = 12, `num_blocks` = $[1, 2, 2]$, `heads` = $[1, 2, 2]$, and `ffn_expansion_factor` = 1.5. This reduction reflects a computational-memory tradeoff: volumetric attention on 3D DWI patches is significantly more expensive than 2D attention, and the reduced capacity is offset by the strong inductive bias from the Hybrid RGS training objective and progressive multi-scale patching.

Third, the downsampling and upsampling operators differ from the orig-

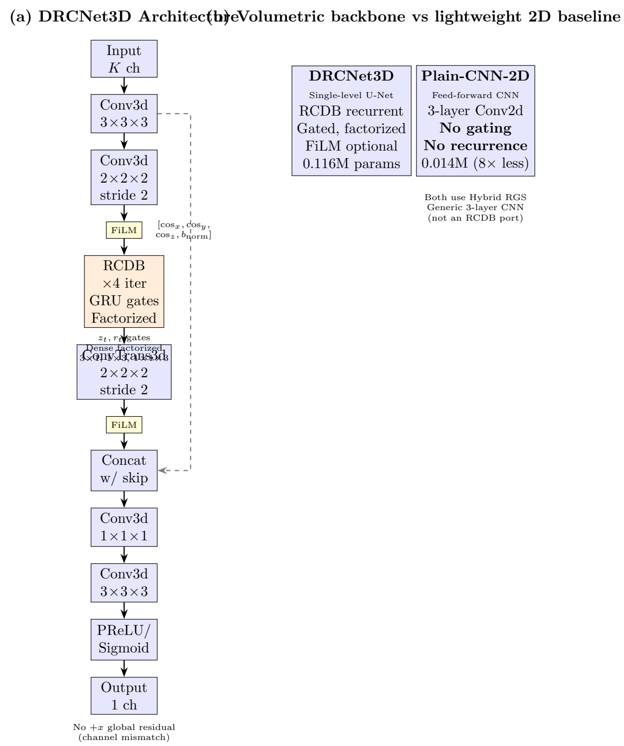

Figure 1: DRCNet architecture for Hybrid RGS. **(a) DRCNet3D internal architecture:** A single-level encoder-decoder with one downsampling and one upsampling stage. The encoder extracts features with Conv3d $3\times3\times3$ and downsamples with strided Conv3d $2\times2\times2$. The core processing uses a Recurrent Convolutional Denoising Block (RCDB) that applies gated recurrent refinement four times, using GRU-style update (z_t) and reset (r_t) gates with dense asymmetric factorized convolutions (parallel $3\times1\times1$, $1\times3\times1$, $1\times1\times3$ kernels fused by $1\times1\times1$). The decoder upsamples with ConvTranspose3d $2\times2\times2$, concatenates with the encoder skip connection, and applies $1\times1\times1$ and $3\times3\times3$ output convolutions followed by PReLU or Sigmoid activation. Optional FiLM conditioning (yellow boxes) modulates features at two stages (bottleneck entry and decoder output) using the target gradient's acquisition metadata $[\cos_x, \cos_y, \cos_z, b_{\text{norm}}]$ (Table 5). Unlike the original DRCNet [2], no global input residual ($+x$) is added due to channel mismatch between K input channels and 1 output channel. **(b) Volumetric backbone versus lightweight 2D baseline:** DRCNet3D preserves the full RCDB recurrent gated architecture with factorized convolutions (0.116M parameters, Table 2). Plain-CNN-2D is an independent generic feed-forward CNN with three Conv2d layers and PReLU activations, with no gating, no recurrence, and no factorized convolutions (0.014M parameters, $8\times$ smaller, Table 3). Note that Plain-CNN-2D is distinct from the original paper's DRCNet-2D variant [2], which preserves the RCDB recurrence in 2D.

inal. The 2D Restormer uses `Conv2d` followed by `PixelUnshuffle(2)` for downsampling and `PixelShuffle(2)` for upsampling, which are parameter-light and lossless channel-space rearrangements. Because PyTorch lacks native 3D pixel-(un)shuffle primitives, the 3D port instead uses strided `Conv3d` for downsampling and `ConvTranspose3d` for upsampling. These learnable resampling layers are less parameter-efficient but provide additional representational flexibility.

Fourth, the input/output stem is redesigned for the Hybrid RGS task. The original Restormer processes RGB images with $\texttt{inp\_channels} = \texttt{out\_channels} = 3$ (or 6 for dual-pixel defocus deblurring) and adds a global input residual connection at the output. In contrast, Restormer-hybrid-RGS uses $\texttt{inp\_channels} = K$ (context gradients plus Bernoulli-masked target at channel index $K-1$) and $\texttt{out\_channels} = 1$ (single denoised target volume). The channel mismatch precludes a direct input residual, so the output head instead applies a learned global affine transformation $(\gamma \cdot \hat{y} + \beta)$ followed by a `PReLU` or `Sigmoid` activation to bound the output range. These trainable scale and shift parameters allow the model to adapt its output distribution during training.

Fifth, an optional gradient-direction conditioning mechanism via Feature-wise Linear Modulation (FiLM) [12] is evaluated as an architectural extension. FiLM layers are inserted at three intermediate stages: after encoder level 2, at the latent bottleneck, and after decoder level 2. Each layer predicts channel-wise scale γ and shift β from the target gradient's four-dimensional acquisition vector $[\cos_x, \cos_y, \cos_z, b_{\text{norm}}]$ via a small multilayer perceptron with identity initialization ($\gamma \approx 1$, $\beta \approx 0$). This conditioning provides weak directional inductive bias without violating the J-invariance of the blind-spot loss. Table 5 reports its effect on D-Brain and Stanford HARDI: FiLM improves full-volume PSNR but slightly degrades ROI PSNR in the current experiments, indicating that it is an exploratory extension rather than a uniformly beneficial component.

The resulting Restormer-hybrid-RGS architecture contains approximately 0.178M parameters (Table 3) and processes 3D patches through the same progressive training schedule as DRCNet-hybrid-RGS. Figure 3 illustrates the Restormer3D architecture and its lightweight 2D baseline counterpart. Because 3D transformer attention is computationally intensive, this model is treated primarily as a scalability and backbone-sensitivity probe to test whether the Hybrid RGS objective transfers to higher-capacity architectures.

3.4.3. Lightweight 2D Baselines for the Capacity-Dimensionality Ablation

The capacity-dimensionality ablation (Section 4.6, Table 3) evaluates whether volumetric processing is necessary for the Hybrid RGS objective or whether lightweight 2D slice-wise backbones suffice. To isolate the contribution of backbone capacity and spatial dimensionality from the self-supervised training framework, we implement two independent generic 2D CNN baselines that process axial slices independently while preserving the same angular gradient sampling and target-channel masking strategy. These models are not dimensionality-reduced ports of the 3D backbones; they are minimal convolutional baselines designed to test whether the RGS self-supervision alone provides effective denoising with reduced architectural complexity.

Plain-CNN-2D is a lightweight feed-forward CNN: a `Conv2d` layer projects the K-channel slice-wise input (context gradients plus Bernoulli-masked target at channel $K-1$) to `base_filters` feature maps, followed by one additional `Conv2d-PReLU` layer and a final `Conv2d` output projection to a single channel. This design has no gating, no recurrence, no factorized convolutions, and no skip connections, in contrast both to the 3D DRCNet-hybrid-RGS backbone described above and to the original DRCNet-2D variant of [2], which preserves the full RCDB recurrence with 2D-factorized convolutions (3×1 and 1×3) and additionally conditions each slice on its four spatial neighbors ($j-2, \ldots, j+2$). Plain-CNN-2D is therefore a minimal convolutional baseline for the ablation (Table 3), not a like-for-like 2D port of the gated recurrent design, and is distinct from the RCDB-based DRCNet-2D of Gurrola et al.

Res-CNN-2D is implemented similarly as a lightweight residual convolutional network. The architecture consists of a `Conv2d` embedding layer, followed by N residual blocks (each comprising two 3×3 `Conv2d` layers with GELU activation and a skip connection), and a final `Conv2d` output projection. This design omits attention mechanisms, hierarchical encoder-decoder structure, and gated feed-forward blocks entirely. The motivation for this simplified baseline is computational: a direct 2D port of the MDTA transformer would be parameter-heavy for slice-wise processing, and the ablation aims to test whether the RGS self-supervision alone is sufficient with minimal architectural capacity. The resulting Res-CNN-2D model has approximately 0.015M parameters, a factor of $12\times$ smaller than the 3D Restormer (Table 3), providing a clear contrast in model complexity while preserving the same training objective.

Both 2D baselines use identical Hybrid RGS training: random gradient subset sampling, Bernoulli target masking with probability $p = 0.3$, masked loss computed only on occluded sites, and the same optimization hyperparameters (Adam, learning rate 4.5×10^{-4}, AMP). The progressive patch schedule follows the same three-stage size sequence (32, 24, 16) and strides (8, 6, 4) as the 3D models, but crops are in-plane $H \times W$ patches (axial slices) rather than cubic 3D volumes; the through-plane dimension is not processed—each axial slice is denoised independently with a K-channel input and no through-plane neighbor conditioning (unlike the original DRCNet-2D of [2], which conditions each slice on four adjacent slices). Batch sizes are matched per stage to equalize memory usage. This ensures that any performance difference observed in the ablation reflects a combination of spatial backbone dimensionality and architectural capacity rather than differences in self-supervised learning strategy or training protocol.

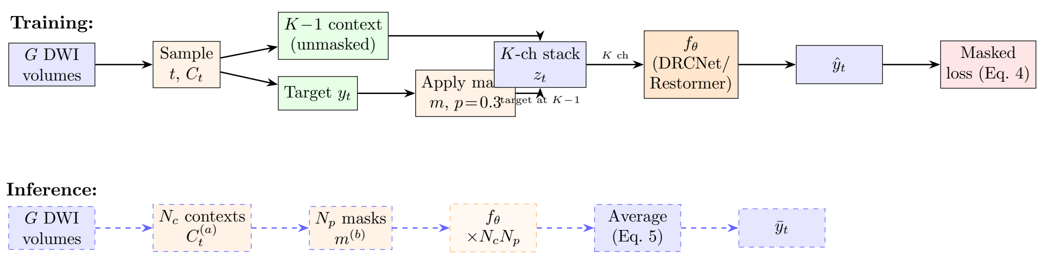

Figure 2: Hybrid RGS training and inference pipeline. **Training (top):** From G diffusion-weighted volumes, a target volume t and context set C_t of $K-1$ gradients are randomly sampled. The context volumes remain unmasked while the target undergoes Bernoulli masking ($p = 0.3$) to create $\tilde{y}_t$. These are concatenated into a K-channel input stack z_t (target at channel $K-1$) and processed by the denoiser f_θ (DRCNet or Restormer). The self-supervised loss (Eq. 4) is computed only on masked voxel locations Ω_m, enforcing a blind-spot objective at supervised masked sites; the J-invariance guarantee applies to the training loss at those locations, not to every voxel of the final reconstruction. **Inference (bottom, dashed):** Each target volume is reconstructed through Monte Carlo ensemble averaging. For N_c random context draws and N_p spatial mask realizations per context, the model generates $N_c \times N_p$ predictions which are averaged to produce the final reconstruction $\bar{y}_t$ (Eq. 5). This ensemble is a stochastic masked ensemble rather than a strictly voxel-wise J-invariant estimator, because each voxel is unmasked in many ensemble members. Default D-Brain configuration: $K = 16$, $N_c = 16$, $N_p = 12$; Stanford HARDI: $N_p = 23$.

3.5. Inference

At inference time, each target volume is reconstructed through a stochastic ensemble procedure (Figure 2, bottom). For each target gradient t, the

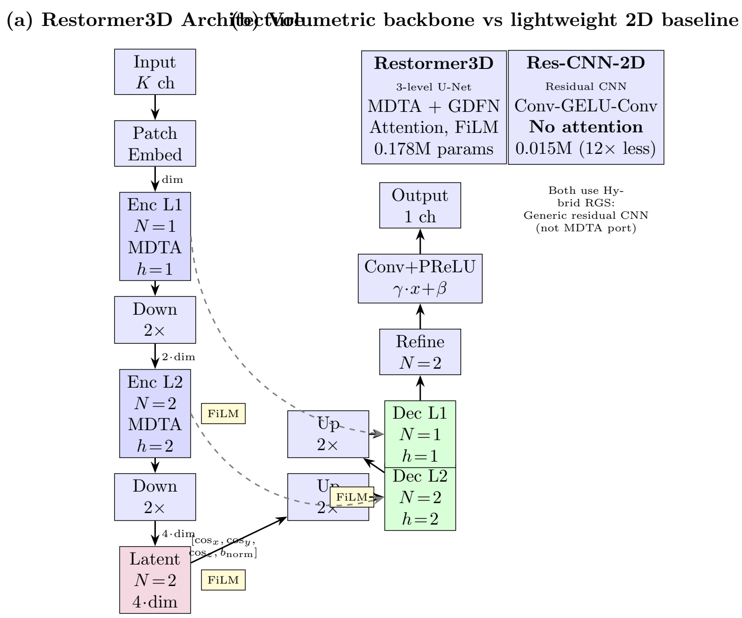

Figure 3: Restormer architecture for Hybrid RGS. **(a) Restormer3D internal architecture:** A 3-level encoder-decoder with transformer blocks at each stage. The encoder progressively downsamples spatial dimensions ($\times 2$ per level) while doubling channel count (dim $\rightarrow$ 2·dim $\rightarrow$ 4·dim, with dim = 12). Each level contains N transformer blocks combining Multi-DConv Head Transposed Self-Attention (MDTA, h heads) and Gated-DConv Feed-Forward Networks (GDFN). The decoder mirrors the encoder with skip connections (dashed gray arrows) and channel-reduction convolutions after concatenation. Refinement blocks and a learned affine output head ($\gamma \cdot x+\beta$) produce the single-channel denoised volume. Optional FiLM conditioning (yellow boxes) modulates features at three stages using the target gradient's acquisition metadata $[\cos_x, \cos_y, \cos_z, b_{\text{norm}}]$; see Table 5 for evaluation. **(b) Volumetric backbone versus lightweight 2D baseline:** Restormer3D is a full 3D transformer with hierarchical attention (0.178M parameters). Res-CNN-2D is an independent generic residual CNN with no attention, no hierarchical structure, and $12\times$ fewer parameters (0.015M), used to test whether the Hybrid RGS objective alone suffices without transformer capacity (Table 3).

method samples N_c random context sets $C_t^{(1)}, \ldots, C_t^{(N_c)}$ from the non-target gradients. For each context, N_p spatial masks are drawn independently, forming masked target inputs and producing N_p full-volume predictions. The final reconstruction averages all predictions:

$$\bar{y}_t = \frac{1}{N_c N_p} \sum_{a=1}^{N_c} \sum_{b=1}^{N_p} f_\theta \left(z_t^{(a,b)} \right), \tag{5}$$

where $z_t^{(a,b)}$ denotes the input formed from context $C_t^{(a)}$ and spatial mask $m^{(a,b)}$, and f_θ is the trained denoiser. This procedure is equivalent to Monte Carlo ensembling over both the angular sampling dimension, through random gradient subsets, and the spatial masking dimension, through Bernoulli occlusion of the target channel. Unlike a masked-voxel-only estimator, which would preserve strict per-voxel J-invariance at test time, this implementation averages complete spatial predictions. The result is a lower-variance reconstruction, but the ensemble can include predictions at unmasked spatial sites and therefore should be interpreted as a stochastic masked ensemble rather than as the strictest voxel-wise J-invariant estimator.

Unless otherwise stated, the D-Brain experiments use $N_c = 16$ context samples and $N_p = 12$ spatial mask draws per context, totaling 192 forward passes per target volume. Stanford HARDI uses $N_c = 16$ and $N_p = 23$ because of the larger gradient shell with $G = 150$ diffusion-weighted directions. The inference cost therefore scales as $O(G N_c N_p)$ forward passes for a full acquisition.

4. Experiments

4.1. Datasets and Protocols

The primary experiments use D-Brain and Stanford HARDI data. For D-Brain, synthetic Rician noise levels are added when clean references are available, allowing direct PSNR, SSIM, and tensor-metric comparisons. Stanford HARDI is used to test behavior on real scanner data and protocol transfer without clean ground truth.

For synthetic D-Brain experiments, noise is added after per-volume min–max normalization. For each voxel with clean normalized signal $s \in [0, 1]$ and noise level σ, the corrupted magnitude is

$$y = \sqrt{(s + \eta_1)^2 + \eta_2^2}, \qquad \eta_1, \eta_2 \sim \mathcal{N}(0, \sigma^2), \tag{6}$$

with independent Gaussian draws for every voxel and diffusion volume, where σ denotes the standard deviation. The result is clipped to $[0, 1]$. This per-voxel, per-volume model matches the J-invariant independence assumptions across masked spatial sites and gradients. Noise robustness uses $\sigma \in \{0.05, 0.10, 0.15, 0.20\}$, with $\sigma = 0.10$ as the primary condition.

Image-domain evaluation reports MSE, PSNR, and SSIM on full 4D arrays and inside a brain-tissue region of interest (ROI). Full-image metrics use every spatial voxel and diffusion channel. PSNR uses the clean reference peak intensity as the dynamic range,

$$\mathrm{PSNR} = 20 \log_{10} \left(\frac{\max(X)}{\sqrt{\mathrm{MSE}}} \right), \tag{7}$$

where X is the original clean array. For SSIM, `structural_similarity` from `skimage.metrics` is called with `data_range=max-min`, `channel_axis=3` for diffusion volumes, and default local-window settings [6]. The ROI mask is $\Omega_{\mathrm{ROI}} = \{i : \max_g Y_i(g) > 0.02\}$, computed from the clean reference array and defined independently of the stochastic Bernoulli training mask m. MSE-ROI and PSNR-ROI use only ROI voxels (Ω_{ROI}); PSNR-ROI uses the same clean-reference dynamic range as full-volume PSNR, i.e., $\max(X)$ over the full clean array. SSIM-ROI is computed on the ROI bounding-box crop to preserve local neighborhoods.

Tensor-derived biomarkers are computed with DIPY 1.x `TensorModel` using ordinary least squares and no additional regularization [7]. Before fitting, denoised DWI volumes are denormalized to the original phantom scale using stored per-volume min–max parameters. The gradient table is built from acquisition b-values and b-vectors, and $b0$ images ($b \leq 50$ s/mm^2) are prepended to the denoised DWI stack.

For **D-Brain**, FA, MD, AD, and RD maps are computed for the denoised output and compared against the clean phantom ground truth using ROI mean absolute errors. For example,

$$\text{FA-MAE} = \frac{1}{|\Omega_{\mathrm{ROI}}|} \sum_{i \in \Omega_{\mathrm{ROI}}} |\, \mathrm{FA}_{\mathrm{denoised},i} - \mathrm{FA}_{\mathrm{GT},i} \,|, \tag{8}$$

with analogous MD-MAE definitions reported in the main tables. FA is dimensionless. MD errors use the units returned by DIPY after denormalization; for D-Brain these are relative diffusivity magnitudes from the phantom scale and should be compared across methods, not interpreted as physical diffusivities in mm^2/s.

For **Stanford HARDI**, no clean ground truth exists. Tensor maps are computed from the denoised output and, where reported, compared against an internal reference derived from the same noisy acquisition (e.g., the tensor fit of the acquired signal before denoising). Such comparisons measure deviation from the input-signal tensor fit rather than ground-truth error, and the term "MAE" is avoided for Stanford; results are described as deviations from the internal reference estimate. The software components are `src/utils/noise.py` for Rician corruption, `src/utils/metrics.py` for PSNR/SSIM, and `src/paper_eval/dti_metrics.py` for tensor fitting.

4.2. Baselines

The proposed framework is compared against the following baselines:

- noisy input without denoising;
- MP-PCA denoising as a classical diffusion MRI baseline;
- Patch2Self with the DIPY OLS backend;
- Patch2Self with the repository reference implementation when applicable;
- MD-S2S or Self2Self-style masking models;
- two-dimensional slice-wise neural denoising variants;
- supervised reference models trained with clean targets or synthetic paired data.

4.3. Evaluation Metrics

Image-domain quality is assessed using MSE, PSNR, and SSIM over the full image and within brain masks or tissue ROIs. Diffusion-specific preservation is assessed through errors in FA, MD, AD, and RD after tensor fitting. Runtime and memory usage are reported for each method to quantify practical deployment cost.

4.4. Ablation Studies

The ablation plan is designed to isolate the main design choices of the framework:

- objective-controlled DRCNet variants separating angular-only, spatial-only, sequential hybrid, and full Hybrid RGS training;
- sequential context selection versus random gradient subset sampling with the same K;
- variation of K, the number of gradients per sampled input group;
- backbone sensitivity across DRCNet-hybrid-RGS, Restormer-hybrid-RGS, and lightweight 2D baselines;
- backbone capacity and spatial dimensionality: volumetric 3D architectures versus lightweight 2D slice-wise baselines under the same Hybrid RGS objective.

The masking probability $p = 0.3$, the number of inference context samples $N_c = 16$, the number of inference mask draws $N_p = 12$, and the progressive patch schedule are fixed implementation defaults in all reported experiments and are not swept in this manuscript.

4.5. Objective-Controlled Ablation

The main comparison in Table 7 motivates this ablation because the Hybrid RGS instantiations improve substantially over Patch2Self and MD-S2S in image-domain ROI metrics. However, that comparison alone does not determine whether the gain comes from the DRCNet backbone or from the self-supervised objective. We therefore perform an objective-controlled ablation in which the backbone is fixed to the same 3D DRCNet and only the training objective is varied. The ablation separates three design axes: whether the model receives angular context from non-target diffusion gradients, whether the target channel is spatially masked to enforce a blind-spot loss, and whether the angular context is sampled randomly rather than deterministically. All four conditions use the same D-Brain data with Rician noise at $\sigma = 0.1$, normalization, patching strategy, optimizer, learning rate, training budget, and evaluation protocol.

The first condition, DRCNet Angular-Only, is a neural Patch2Self-style objective. The network receives $K - 1$ randomly sampled context gradients that exclude the target volume and predicts the full target volume with a full-volume loss. Because the target image is absent from the input, no spatial target masking is required. This condition tests whether angular redundancy alone is sufficient when the linear Patch2Self regressor is replaced by the same nonlinear DRCNet backbone. The second condition, DRCNet Spatial-Only, is a volumetric Self2Self-style objective. The network receives only the target volume with Bernoulli-masked voxels and is trained with a masked loss, with no auxiliary diffusion gradients. This condition tests whether local spatial context alone can denoise DWI volumes without angular information. The third condition, DRCNet Sequential Hybrid, combines a masked target channel with $K-1$ auxiliary gradients, but the auxiliary gradients are selected by a deterministic sequential window. The fourth condition is the proposed DRCNet Hybrid RGS objective, which combines the masked target channel with $K-1$ randomly sampled non-target gradients and computes the loss only on masked voxels.

Table 1 shows that angular context alone is not sufficient in this setting. DRCNet Angular-Only reaches only 10.87 dB PSNR-ROI, despite using the same neural backbone. This failure is informative rather than merely negative: when the target volume is completely excluded from the input, the network must infer each voxel entirely from other diffusion directions. Those directions are correlated with the target but do not contain the target's high-frequency spatial detail, leading to severe image-domain degradation. This result rules out the narrow explanation that replacing Patch2Self's linear regression with a nonlinear DRCNet under the same limited random-context channel budget is enough to obtain the Hybrid RGS gains.

The DRCNet Angular-Only row in Table 1 uses $K-1=15$ randomly sampled context gradients as input and contains no target channel. It should therefore be interpreted as a matched-channel-budget ablation, not as a full neural replacement for Patch2Self. The purpose is to isolate the contribution of target-channel spatial masking while keeping the input dimensionality comparable to Hybrid RGS with $K=16$. In contrast, a full neural Patch2Self-style model would condition on all $G-1=59$ non-target gradients and would require a different input layer, memory budget, and training configuration; this comparison is left for future work. The restricted channel budget helps explain the degradation to 10.87 dB PSNR-ROI: inferring each target voxel from only 15 randomly selected diffusion directions, without the target volume's own spatial context, does not recover high-frequency anatomical detail.

Spatial masking alone is much stronger. DRCNet Spatial-Only reaches 24.47 dB PSNR-ROI and the best FA-MAE in this ablation, showing that the blind-spot target-channel objective captures substantial local anatomical structure. It also outperforms DRCNet Sequential Hybrid in PSNR-ROI (24.47 versus 23.60 dB), which suggests that adding angular context through a fixed deterministic window can introduce bias or unstable channel dependencies if the angular neighborhoods are not sufficiently diverse. However, Spatial-Only remains more than 2 dB below full Hybrid RGS, indicating that angular context is useful when it is coupled with appropriate sampling.

The full DRCNet Hybrid RGS objective achieves the highest image-domain fidelity, with 26.93 dB PSNR-ROI and 0.5247 SSIM-ROI. The sequential-versus-random comparison in Table 2 already suggested the importance of random angular context; this objective-controlled grouping makes the interpretation clearer. With the architecture fixed, the gain over Sequential Hybrid indicates that random subsets act as angular data augmentation, exposing the model to varied diffusion contexts and discouraging overfitting to fixed channel neighborhoods. FA-MAE and MD-MAE vary over a narrower range than PSNR-ROI and SSIM-ROI, so this ablation should not be read as proving large tensor-domain superiority. Instead, the main conclusion is that, using the same DRCNet backbone, Hybrid RGS outperforms angular-only, spatial-only, and sequential hybrid objectives in image reconstruction fidelity, confirming that the improvement stems from the coupled random-gradient-subset and target-masking training scheme rather than from network capacity alone.

Table 1: Objective-controlled ablation on D-Brain with Rician noise $\sigma = 0.1$. All rows use the same DRCNet backbone, training budget, and evaluation protocol. The ablation isolates angular context, target-channel masking, and random gradient subset sampling.

Training objective	Angular	Masking	Random	PSNR-ROI ↑	SSIM-ROI ↑	FA-MAE ↓	MD-MAE† ↓	Time/vol. (s)
DRCNet Angular-Only	✓	–	✓	10.87	0.3407	0.2328	3499	5.2
DRCNet Spatial-Only	–	✓	–	24.47	0.4778	**0.2234**	**3457**	4.5
DRCNet Sequential Hybrid	✓	✓	–	23.60	0.4751	0.2390	3555	**3.7**
DRCNet Hybrid RGS	✓	✓	✓	**26.93**	**0.5247**	0.2575	3451	34.1

Note: †MD-MAE values are reported in arbitrary units derived from the D-Brain phantom intensity scale and should be interpreted as relative comparisons across methods.

4.6. Sampling Strategy Ablations

Hybrid RGS relies on random gradient subsets rather than deterministic angular windows. To validate this choice, we first compare random gradient sampling against a sequential strategy with the same number of input channels. In RGS mode, each training step samples K distinct gradient indices uniformly from the full shell of G diffusion directions and shuffles the resulting stack, with the target assigned to the fixed prediction channel. In sequential mode, the network instead sees deterministic sliding windows of adjacent gradients. This experiment asks whether randomization provides useful angular data augmentation or whether the blind-spot objective alone is sufficient when the channel count is fixed.

Both sampling modes are evaluated on D-Brain with Rician noise at $\sigma = 0.1$ using $K = 16$, the same masking probability, learning rate, patching strategy, and number of epochs. Sequential mode constructs windows [0:16], [1:17], ..., $[G - K{:}G]$, with the last channel treated as the prediction target. RGS mode samples 16 distinct indices without replacement in each batch and places the target at channel 15. We evaluate both DRCNet and Restormer to test whether the benefit of random angular context is architecture dependent.

The results in Table 2 show that random sampling consistently improves over deterministic windows. For DRCNet, RGS improves PSNR-ROI from 23.60 to 26.93 dB and yields comparable FA-MAE. For Restormer, RGS improves PSNR-ROI from 21.64 to 23.43 dB, with a small increase in FA-MAE. The improvement is not interpreted as a surprising failure of sequential windowing: adjacent gradients are still informative and the J-invariant masking objective remains active in both modes. Rather, the consistent gap supports the view that RGS acts as angular data augmentation. By exposing the model to many non-local combinations of diffusion directions, it discourages memorization of local angular neighborhoods and encourages reconstruction from more diverse diffusion contexts.

A second ablation studies the number of input volumes K. Too small a value limits angular information and may force the network to rely primarily on spatial interpolation. Too large a value increases memory, parameter count, and runtime while potentially providing redundant diffusion measurements. Establishing a practical K is important because acquisition protocols vary substantially: D-Brain uses $G = 60$ diffusion volumes, whereas larger HARDI protocols can exceed $G = 100$.

We evaluate $K \in \{5, 10, 16, 24, 30\}$ on D-Brain under the same noise setting. Each value of K requires a separate model because the input channel dimension changes. DRCNet is evaluated across the full sweep, and Restormer is evaluated with the same values to check whether the trend persists for the transformer backbone. All runs use the same core Hybrid RGS protocol, including the same number of context samples and stochastic predictions at inference when available.

The quality curves show a plateau rather than monotonic improvement. DRCNet obtains similar PSNR-ROI at $K = 5$ and $K = 10$ (26.46 dB), reaches 26.93 dB at the recommended $K = 16$ configuration, and changes only marginally at larger values (26.92 dB for $K = 24$ and 26.36 dB for $K = 30$). Restormer shows its strongest PSNR-ROI at $K = 10$ (26.60 dB), while $K = 16$ is lower in the current run (23.43 dB) and larger values do not recover a clear advantage. This indicates that the benefit of adding more diffusion volumes saturates quickly and can become architecture dependent.

The runtime and model-size trends favor moderate K. For DRCNet, the parameter count grows from 0.106M at $K = 5$ to 0.128M at $K = 30$, while inference time increases from 32.5 to 37.2 s per target volume. Restormer is more expensive overall, ranging from 129.2 to 131.9 s per volume across the same sweep. Although the inference-time increase is modest in these runs, larger K also increases memory pressure and training cost. We therefore use $K = 16$ as the recommended configuration: it captures most of the available angular benefit, supports both backbones without excessive input dimensionality, and leaves room for larger K only when protocols have many gradients and compute budget is less restrictive.

Figure 4 visualizes the same tradeoff as a quality-versus-input-size curve. The key practical conclusion is that Hybrid RGS does not require using a large fraction of all diffusion volumes at once. Moderate subsets provide sufficient angular context for denoising while preserving feasible 3D training and inference.

4.7. Backbone Capacity and Spatial Dimensionality

A critical question for Hybrid RGS is whether its gains come primarily from the self-supervised sampling-and-masking scheme or from the use of volumetric neural backbones. Many prior self-supervised denoising methods process medical images slice by slice with two-dimensional convolutions, including Noise2Void-style adaptations and MD-S2S-like masking strategies. In contrast, the proposed models are designed to operate on 3D patches so that they can exploit inter-slice anatomical continuity. This ablation isolates

Table 2: Sampling ablations on D-Brain with Rician noise $\sigma = 0.1$. The first block compares sequential windows and random gradient sampling at $K = 16$. The second block summarizes the K sweep. Params and time/vol. are taken from the corresponding runtime registry when available.

Architecture	Sampling / K	PSNR-ROI ↑	FA-MAE ↓	Params (M)	Time/vol. (s)
DRCNet	Sequential, $K = 16$	23.60	0.2390	0.116	3.7
DRCNet	RGS, $K = 16$	**26.93**	0.2575	0.116	34.1
Restormer	Sequential, $K = 16$	21.64	0.2465	0.178	104.2
Restormer	RGS, $K = 16$	**23.43**	0.2238	0.178	126.8
DRCNet	RGS, $K = 5$	26.46	0.2319	0.106	32.5
DRCNet	RGS, $K = 10$	26.46	0.2378	0.111	33.8
DRCNet	RGS, $K = 16$	**26.93**	0.2575	0.116	34.1
DRCNet	RGS, $K = 24$	26.92	0.2559	0.123	36.1
DRCNet	RGS, $K = 30$	26.36	0.2464	0.128	37.2
Restormer	RGS, $K = 5$	25.88	0.2489	0.174	129.2
Restormer	RGS, $K = 10$	**26.60**	**0.2195**	0.176	129.2
Restormer	RGS, $K = 16$	23.43	0.2238	0.178	126.8
Restormer	RGS, $K = 24$	25.83	0.2198	0.180	130.2
Restormer	RGS, $K = 30$	25.50	0.2199	0.182	131.9

the architectural contribution by keeping the Hybrid RGS training objective fixed while varying both spatial dimensionality and backbone capacity. The resulting comparison is important for interpretation: if lightweight 2D baselines remain competitive, then the principal contribution is the RGS self-supervision scheme; if the 3D volumetric backbones improve substantially, then higher capacity and volumetric context are essential parts of the method. It is important to note that this comparison confounds dimensionality with architectural family and capacity, so the results reflect the combined effect of these factors rather than a pure test of 2D versus 3D convolutions.

The 3D configuration corresponds to the proposed setup: a $K = 16$ RGS input stack, Bernoulli masking on the target channel, and a 3D DRCNet or 3D Restormer backbone trained on volumetric patches from D-Brain with Rician noise at $\sigma = 0.1$. The 2D baseline configuration preserves the same angular sampling, target-channel masking, loss function, learning rate, and training protocol, but replaces volumetric operations with slice-wise processing using lightweight generic CNN architectures. Each axial slice is denoised independently with a K-channel input, so the model can exploit angular information across diffusion gradients but has no explicit convolutional or attention connectivity along the through-plane direction. Plain-CNN-

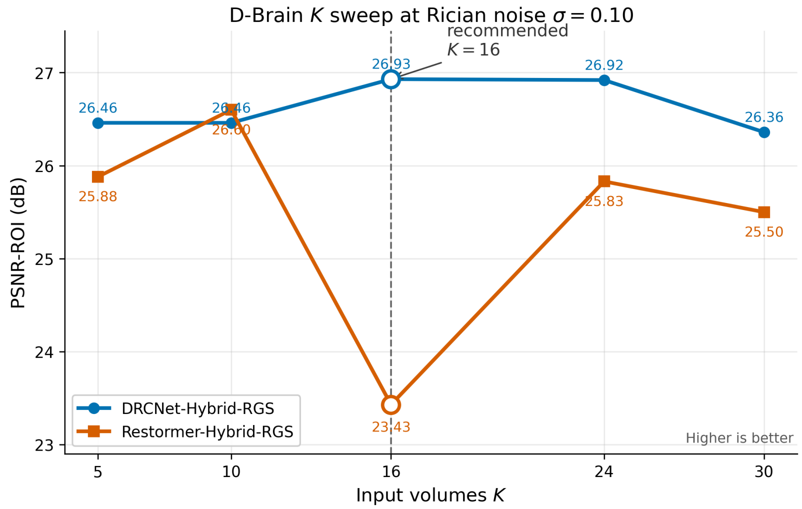

Figure 4: K-sweep visualization on D-Brain. The plot shows PSNR-ROI versus $K \in \{5, 10, 16, 24, 30\}$ for DRCNet and Restormer, with the recommended $K = 16$ configuration annotated.

2D is a minimal feed-forward convolutional network, and Res-CNN-2D is a lightweight residual CNN variant (see Section 3.4), neither of which are architectural ports of the 3D backbones.

Table 3 shows that the current controlled ablation does not support a simple claim that 3D processing is uniformly superior. All comparisons in this table use the fixed $K = 16$ operating point. For the DRCNet backbone family, Plain-CNN-2D reaches 27.11 dB PSNR-ROI and 0.5569 SSIM-ROI, compared with 26.93 dB and 0.5247 for DRCNet3D. Plain-CNN-2D also obtains lower FA-MAE (0.2296 versus 0.2575) and slightly lower MD-MAE (3435 versus 3451). For the Restormer backbone family, Res-CNN-2D reaches 26.44 dB PSNR-ROI, whereas Restormer3D reaches 23.43 dB at $K = 16$. Res-CNN-2D improves PSNR-ROI, SSIM-ROI, and MD-MAE, while Restormer3D retains a slight FA-MAE advantage (0.2238 versus 0.2309). It is important to note that this comparison is not representative of Restormer's best configuration: the K sweep (Table 2) shows Restormer at $K = 10$ reaching PSNR-ROI = 26.60 dB, which exceeds Res-CNN-2D (26.44 dB). The conclusion that lightweight 2D baselines outperform volumetric Restormer therefore holds specifically at $K = 16$ and should not be generalized. These results suggest that, under the current training and inference settings at the fixed operating point, the Hybrid RGS objective itself provides a large fraction of the denoising benefit, and the larger 3D architectures do not automatically improve most quantitative image-domain or tensor metrics.

This outcome is informative rather than problematic. The 2D baselines still use the same Hybrid RGS mechanism, so their strong performance validates the algorithmic contribution: random angular context and blind-spot target masking are sufficient to train effective self-supervised denoisers even with minimal architectural capacity. The absence of a clear 3D advantage may reflect the anisotropy of the acquisition, the patch size and masking schedule, or the fact that adjacent axial slices add less independent information than additional diffusion directions. It may also indicate that the current metrics, computed volume-wise and through tensor summaries, do not fully capture slice-to-slice consistency. Because the 2D baselines differ substantially in capacity and architecture family from their 3D counterparts, this comparison should be interpreted as a combined test of backbone capacity and spatial dimensionality rather than a pure dimensionality ablation. Qualitative inspection of reconstructed volumes, FA maps, and tractography-derived summaries remains necessary to determine whether the higher-capacity 3D models reduce through-plane discontinuities that are not penalized strongly by PSNR or voxelwise tensor errors.

The computational comparison further favors the 2D baselines for the present implementation. Plain-CNN-2D uses approximately 0.014M parameters and requires 4.7 s per target volume, whereas DRCNet3D uses 0.116M parameters and requires approximately 34.1 s. Res-CNN-2D shows an even larger gap: 0.015M parameters and 3.9 s for the 2D baseline versus 0.178M parameters and 126.8 s for Restormer3D. Therefore, the current evidence supports a nuanced conclusion: Hybrid RGS is architecture-agnostic and works well even with lightweight 2D backbones, while higher-capacity 3D processing should be justified by qualitative volumetric coherence or downstream tractography benefits rather than by the present scalar metrics alone.

Note: †Diffusivity errors are in arbitrary D-Brain phantom units; compare across methods, not as physical mm^2/s values.

4.8. Robustness to Noise Level

Real-world DW-MRI data exhibit variable noise levels because of scanner hardware, field strength, coil sensitivity, acquisition time, acceleration, and patient motion. A practical denoising method should therefore behave predictably across signal-to-noise regimes rather than being tuned to a single

Table 3: Backbone capacity and spatial dimensionality ablation on D-Brain with Rician noise $\sigma = 0.1$ and $K = 16$. All backbones use the same Hybrid RGS sampling and masked self-supervised loss. The comparison confounds dimensionality with architectural family and capacity: the 2D baselines are lightweight generic CNNs, not architectural ports of the 3D backbones.

Backbone	Processing	PSNR-ROI ↑	SSIM-ROI ↑	FA-MAE ↓	MD-MAE† ↓	Params (M)	Time/vol. (s)
DRCNet3D	Volumetric	26.93	0.5247	0.2575	3451	0.116	34.1
Plain-CNN-2D	Slice-wise	**27.11**	**0.5569**	**0.2296**	**3435**	0.014	**4.7**
Restormer3D	Volumetric	23.43	0.5107	**0.2238**	3440	0.178	126.8
Res-CNN-2D	Slice-wise	**26.44**	**0.5504**	0.2309	**3432**	0.015	**3.9**

synthetic noise level. We evaluate this property by varying the Rician noise level on D-Brain and asking whether Hybrid RGS degrades smoothly at high noise while preserving diffusion-derived quantities. This experiment is not a cross-noise transfer study: each model is trained and evaluated at the same noise level, isolating robustness of the training protocol rather than domain adaptation across noise distributions.

We evaluate $\sigma \in \{0.05, 0.10, 0.15, 0.20\}$ with the same D-Brain ground-truth reference and the same Hybrid RGS configuration ($K = 16$). For the proposed methods, DRCNet-Hybrid-RGS and Restormer-Hybrid-RGS are trained separately for each σ. MP-PCA is evaluated at each noise level as a classical diffusion MRI baseline. Patch2Self and MD-S2S are available for the lower-noise settings in the current registry and are included where present. We report PSNR-ROI as the primary image-domain metric and FA-MAE as a downstream tensor-preservation metric. Full SSIM, MSE, MD, AD, and RD results are left to supplementary tables. The $\sigma = 0.10$, $K = 16$ entries correspond to the canonical primary-condition runs used for manuscript reporting; separate ablation-family runs are used only within their corresponding ablation tables.

Table 4 and Figure 5 show an overall decline in PSNR-ROI as the noise level increases, with small non-monotonic variations for some methods in the intermediate-noise regime. MP-PCA is particularly strong in this controlled synthetic setting, achieving the highest PSNR-ROI at $\sigma = 0.05$ (33.33 dB) and remaining marginally ahead of DRCNet at $\sigma = 0.20$ (23.55 versus 23.52 dB). This result should not be hidden: local PCA assumptions are well matched to spatially homogeneous synthetic Rician corruption, and MP-PCA remains a highly competitive preprocessing method for diffusion MRI. The proposed neural methods are nevertheless stable across the sweep. DRCNet decreases from 29.76 dB at $\sigma = 0.05$ to 23.52 dB at $\sigma = 0.20$, while Restormer decreases from 27.19 to 23.02 dB. Neither architecture shows catastrophic failure at the highest noise level.

The low-noise regime illustrates the tradeoff between classical denoising and learned self-supervision. At $\sigma = 0.05$, MP-PCA gives the best PSNR-ROI and FA-MAE, while DRCNet-Hybrid-RGS still substantially outperforms Patch2Self (29.76 versus 23.74 dB) and MD-S2S (16.77 dB) in image-domain fidelity. Restormer-Hybrid-RGS is lower than DRCNet in PSNR-ROI but obtains a stronger FA-MAE than DRCNet (0.2170 versus 0.2579). Thus, even when the noise level is mild, different methods occupy different points in the image-fidelity versus tensor-preservation tradeoff.

At higher noise levels, the learned models remain competitive with MP-PCA in PSNR-ROI but do not dominate FA-MAE. At $\sigma = 0.15$, DRCNet reaches 25.11 dB compared with 25.54 dB for MP-PCA; at $\sigma = 0.20$, the two are nearly tied. However, MP-PCA retains much lower FA-MAE across all tested noise levels. In contrast, the neural methods maintain more stable MD errors, suggesting that scalar tensor metrics respond differently to each denoising prior. The practical conclusion is therefore nuanced: Hybrid RGS is robust in the sense of stable training and graceful degradation across SNR regimes, but MP-PCA remains a strong FA-preserving baseline under this synthetic noise model. This motivates reporting both image-domain and diffusion-domain metrics rather than relying on PSNR alone.

Table 4: Noise-level robustness on D-Brain. Values are PSNR-ROI (dB); FA-MAE is shown in parentheses for the proposed methods and MP-PCA. Patch2Self and MD-S2S entries are included where available in the current registry. Each neural model is trained and evaluated at the same Rician noise level.

σ	MP-PCA	Patch2Self	MD-S2S	DRCNet-Hybrid-RGS	Restormer-Hybrid-RGS
0.05	**33.33** (0.1009)	23.74	16.77	29.76 (0.2579)	27.19 (0.2170)
0.10	28.44 (0.0898)	21.31	15.28	26.93 (0.2575)	23.43 (0.2238)
0.15	**25.54** (0.0978)	–	–	25.11 (0.2202)	24.63 (0.2452)
0.20	**23.55** (0.0863)	–	–	23.52 (0.2211)	23.02 (0.2597)

4.9. Gradient Direction Conditioning via FiLM

In the baseline Hybrid RGS formulation, the K-channel input stack is randomly sampled and shuffled at each training step. Although this stochasticity is useful for preventing the model from relying on a fixed channel order, it also means that the network receives no explicit signal indicating which

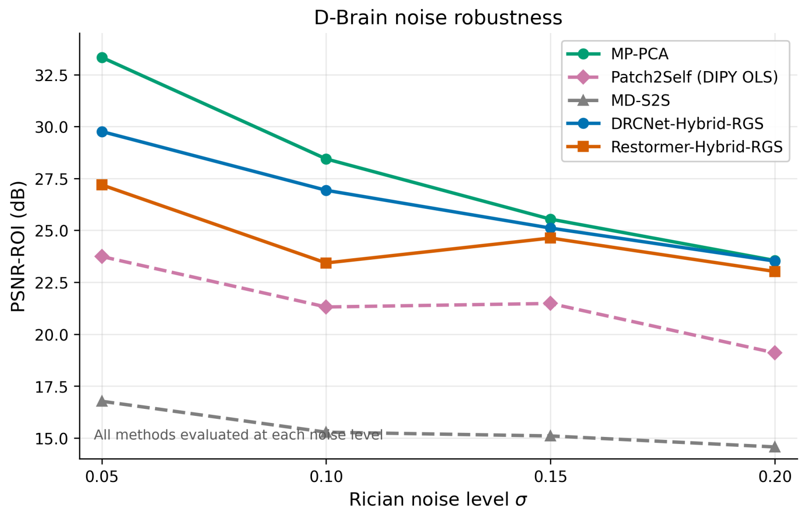

Figure 5: Noise-robustness curve on D-Brain. The plot shows PSNR-ROI versus Rician noise level $\sigma \in \{0.05, 0.10, 0.15, 0.20\}$ for MP-PCA, Patch2Self, MD-S2S, DRCNet-Hybrid-RGS, and Restormer-Hybrid-RGS, with missing baseline points omitted where runs are unavailable.

physical diffusion gradient corresponds to the target channel. A natural question is therefore whether direction-aware conditioning can improve denoising, particularly in anatomically complex regions where the relationship between diffusion signal and local fiber organization is direction dependent. This question is analogous to the role of positional encodings in transformer models: the network can, in principle, infer structure from the data alone, but explicit metadata may make the learning problem easier. We evaluate this idea as an extension of the framework rather than as a required component of Hybrid RGS.

An initial design iteration considered an additive orientation encoding at the input level. In this approach, the metadata vector $[\cos_x, \cos_y, \cos_z, b_{\text{norm}}]$ for each input volume was projected into a learned two-dimensional spatial pattern, interpolated to the patch resolution, expanded along the third spatial dimension, and added to all K input channels before the backbone. We discarded this design on theoretical grounds before treating it as a valid experimental condition. The approach generates artificial spatial patterns from spatially invariant acquisition metadata, risks dilution as the signal propagates through the network, and overconditions all context channels rather than focusing the modulation on the target volume being reconstructed.

We instead use Feature-wise Linear Modulation (FiLM) [12] as a lightweight feature-space conditioning mechanism. Given an intermediate feature tensor F, FiLM applies

$$\mathrm{FiLM}(F) = \gamma \odot F + \beta, \tag{9}$$

where the channel-wise scale γ and shift β are predicted by a small multilayer perceptron from the four-dimensional condition vector of the target channel only. This target-only design matches the RGS objective: the model is informed about the physical direction it must reconstruct, while the randomly sampled context volumes remain auxiliary observations rather than separately conditioned targets.

FiLM differs from the discarded additive encoding in five important ways. First, conditioning is applied only to the target gradient direction, avoiding unnecessary modulation of all K input channels. Second, metadata enters in feature space rather than as an artificial image-like pattern. Third, modulation can be injected at multiple depths, allowing the condition to influence both early local features and deeper semantic representations. Fourth, the conditioning vector remains spatially invariant, consistent with the fact that gradient direction and b-value describe acquisition geometry rather than anatomical location. Fifth, the FiLM layers use identity initialization, with γ initialized near one and β near zero, so the conditioned model starts from behavior close to the unconditioned baseline.

In DRCNet-hybrid-RGS, FiLM is inserted at two feature stages of the 3D convolutional backbone. In Restormer-hybrid-RGS, it is inserted at three feature stages to modulate multiple levels of the transformer-style restoration hierarchy. The resulting parameter overhead is small: approximately 3.9% for DRCNet and 3.8% for Restormer. The mechanism preserves the J-invariant training principle because the conditioning vector is deterministic acquisition metadata and does not reveal the noisy voxel intensities being predicted under the blind-spot loss.

Note: †Diffusivity errors are in arbitrary D-Brain phantom units; compare across methods, not as physical mm^2/s values.

Under the registry summarized in Table 5, FiLM has mixed effects on D-Brain. For DRCNet, conditioning increases full-volume PSNR from 23.90 to 25.40 dB, a gain of 1.51 dB, but slightly decreases ROI PSNR from 26.93 to 26.24 dB (-0.69 dB). The full-volume MSE decreases by 29.3%, whereas ROI MSE increases by 17.2%, indicating that the global image-domain gain is not concentrated in brain tissue. FA-MAE improves by 6.6%, and MD-MAE

Table 5: Gradient direction conditioning ablation on D-Brain. Under the authoritative registry, FiLM improves full-volume PSNR for both backbones but slightly degrades PSNR-ROI, indicating metric-dependent behavior rather than a uniform denoising benefit.

Model	Conditioning	PSNR (dB) ↑	PSNR-ROI (dB) ↑	FA-MAE ↓	MD-MAE† ↓	Params	Time/vol.
DRCNet	Baseline RGS	23.90	26.93	0.2575	3451	116K	34.1 s
DRCNet	FiLM	25.40	26.24	0.2405	3452	120K	35.3 s
DRCNet	Δ FiLM − baseline	+1.51	−0.69	−6.6%	+0.0%	+3.9%	+3.2%
Restormer	Baseline RGS	22.98	23.43	0.2238	3440	178K	126.8 s
Restormer	FiLM	23.60	22.69	0.2603	3423	185K	130.3 s
Restormer	Δ FiLM − baseline	+0.63	−0.74	+16.4%	−0.5%	+3.8%	+2.8%

changes little from 3451 to 3452.

For Restormer, FiLM improves full-volume PSNR by 0.63 dB but decreases ROI PSNR from 23.43 to 22.69 dB (−0.74 dB). The full-volume MSE decreases by 13.4%, whereas ROI MSE increases by 18.4%. MD, AD, and RD errors improve slightly, but FA-MAE worsens by 16.4%, again showing that full-volume image-domain gains do not necessarily translate to brain-ROI or tensor-derived improvements. FiLM should therefore be interpreted as an exploratory conditioning mechanism that requires architecture-specific tuning rather than as a recommended default extension.

The benefit of FiLM is therefore metric and architecture dependent. DRCNet, which has a simpler convolutional inductive bias, shows a larger full-volume PSNR gain and improved FA-MAE, whereas Restormer shows a smaller full-volume PSNR gain and worsened FA-MAE. Both backbones lose ROI PSNR, suggesting that FiLM placement, identity initialization, regularization, or tensor-aware validation may need to be tuned before conditioning can be claimed to improve anatomically relevant regions. On Stanford HARDI, FiLM-trained models converged successfully under a different protocol with $G = 150$ gradients and $b = 2000$ s/mm^2, supporting the feasibility of the conditioning mechanism beyond D-Brain. Because no matching unconditioned Stanford tensor-summary run is available under the same FiLM-evaluation protocol, those results are used only for qualitative FA/MD assessment and are not included in the quantitative ablation table.

4.10. Implementation Details

Training.

All neural models are trained with Adam using a base learning rate of 4.5×10^{-4} and no learning-rate schedule. Automatic mixed precision (AMP) is enabled to reduce memory consumption on the available GPU. D-Brain training

Table 6: Progressive training schedule used for Hybrid RGS models. Patch sizes are cubic 3D patches.

Stage	Patch size	Stride	Batch size (DRCNet)	Batch size (Restormer)	Epochs
1	32^3 voxels	8	128	64	120
2	24^3 voxels	6	256	128	120
3	16^3 voxels	4	256	256	120

uses `seed=91021` and Stanford training uses `seed=91022`, with `reproducible: true` for PyTorch, NumPy, and the dataset-level Hybrid RGS gradient sampler. D-Brain additionally uses `subset_fraction=0.6` for random patch subsampling during training. Exact bitwise reproducibility across GPU devices is nevertheless not expected because cuDNN operations may remain nondeterministic.

Progressive schedule.
DRCNet and Restormer use the same three-stage progressive patch protocol (Table 6), with architecture-specific batch sizes chosen to fit memory. Each stage constructs a new dataset at the specified patch resolution and resets the optimizer and scheduler, but model weights are preserved and continue training from the previous stage. D-Brain uses 120 epochs per stage, for 360 epochs total. Stanford HARDI uses 300 epochs per stage, for 900 epochs total.

Augmentation.
No explicit geometric augmentation, such as flips, rotations, or elastic deformations, is applied. Instead, training stochasticity is provided by random gradient-subset sampling in RGS mode, independent Bernoulli spatial masks for each training sample, and strided patch extraction over many spatial locations. During dataset construction, background patches satisfying $\max(\text{patch}) \leq 0.02$ are discarded.

Normalization.
Each diffusion-weighted volume is independently min–max normalized to $[0, 1]$ before training. Denoised outputs are clipped to the valid range and, when metric computation requires it, rescaled back to $[0, 1]$ per volume using the configured `rescale_to_01: true` and `clip_to_range: true` settings.

Inference.
Full-volume reconstruction uses Monte Carlo averaging over random context subsets and target masks. D-Brain evaluation uses $N_c = 16$ context samples and $N_p = 12$ mask draws per context, while Stanford HARDI uses $N_c = 16$ and $N_p = 23$. Mask predictions are computed in chunks of four to control GPU memory (`pred_chunk_size=4`). Restormer additionally uses overlapping tiled inference with Gaussian weighting, patch size 32, and overlap 1.

Registry provenance.
For reproducibility, the primary $\sigma = 0.10$, $K = 16$ self-supervised baseline runs are identified by `drcnet_dbrain_rgs_final`, `restormer_dbrain_rgs_final`, `drcnet_stanford_rgs_final`, and `restormer_stanford_rgs_final`, recorded in experiment registry `paper_final_k16_rerun` on June 28, 2026 (`tmp/paper_final_k16_rerun_20260628T042410Z/registry.jsonl`). The reported D-Brain metrics use the best-loss checkpoint and the Monte Carlo protocol above ($N_c = 16$, $N_p = 12$ per target volume); Stanford uses $N_c = 16$ and $N_p = 23$. Runs trained specifically for ablation families, such as the K sweep, are reported only within those ablation contexts.

Hardware and timing.
All experiments are run on a single NVIDIA Tesla T4 GPU with 15 GB of VRAM, CUDA 12.2, and PyTorch 2.x. A complete 360-epoch D-Brain training run requires approximately 8–12 hours depending on the backbone. With the Monte Carlo settings above, inference time per target diffusion volume ranges from approximately 4.7 s for Plain-CNN-2D to 127 s for Restormer3D.

Checkpoint selection and code.
For evaluation, the final-stage checkpoint with the lowest epoch-averaged training loss is retained as `best_loss_checkpoint.pth`. Validation-set early stopping is not used because clean validation targets are unavailable in the self-supervised setting. Training and inference code are provided in `src/drcnet_hybrid_rgs/` and `src/restormer_hybrid_rgs/`, with all hyperparameters specified in the corresponding YAML configuration files. Results should be numerically stable within normal run-to-run variance, but exact bitwise agreement across GPU types is not guaranteed because cuDNN operations remain nondeterministic.

5. Experimental Results

5.1. Main Comparison

Table 7 summarizes the primary quantitative evaluation on the D-Brain dataset with synthetic Rician noise at $\sigma = 0.1$. Hybrid RGS methods use $K = 16$ input gradient volumes as a fixed operating point for cross-model comparability. This value is DRCNet-favorable: as shown in Table 2, DRCNet peaks at $K = 16$ (26.93 dB PSNR-ROI), whereas Restormer's strongest result in the K sweep is at $K = 10$ (PSNR-ROI = 26.60 dB, FA-MAE = 0.2195). The Restormer row in this table therefore reflects a fixed operating point, not Restormer's best-sweep configuration. The table contrasts the proposed self-supervised 3D models against classical and self-supervised baselines: MP-PCA [8], Patch2Self with the DIPY OLS backend [9], and MD-S2S with a two-dimensional convolutional masking strategy [11]. Supervised DRCNet and Restormer variants are included as supervised reference runs to contextualize the self-supervised results; they are not intended as rigorously tuned upper bounds.

MP-PCA remains a strong classical preprocessing baseline. It achieves 28.44 dB PSNR-ROI and the lowest FA-MAE in this experiment (0.0898), which indicates that its local PCA assumptions can preserve tensor anisotropy well under the tested synthetic noise model. However, its MD-MAE remains high (4650), suggesting that improved FA preservation does not necessarily imply uniform improvement across all diffusion tensor scalars. Patch2Self provides a self-supervised angular baseline using leave-one-volume-out OLS regression. Its PSNR-ROI is lower (21.31 dB), but its MD-MAE (3438) is competitive with the neural self-supervised methods, consistent with the strength of angular redundancy in DWI. MD-S2S obtains the lowest FA-MAE among the self-supervised baselines (0.2298) but has poor image-domain fidelity, with 15.28 dB PSNR-ROI, likely because it processes two-dimensional slices independently and does not exploit inter-slice anatomical context.

The proposed Hybrid RGS models combine the angular self-supervision of Patch2Self with the blind-spot spatial masking of Self2Self-style methods. DRCNet-Hybrid-RGS achieves the strongest self-supervised image-domain results in this table: 23.90 dB PSNR, 0.4402 SSIM, and 26.93 dB PSNR-ROI. Relative to Patch2Self, this corresponds to a 5.62 dB gain in PSNR-ROI and approximately a 73% reduction in ROI MSE using the rounded table values. Relative to MD-S2S, DRCNet-Hybrid-RGS improves PSNR-ROI by 11.65 dB, supporting the value of coupling random angular context with target-channel masking. Its FA-MAE (0.2575) is slightly worse than Patch2Self and MD-S2S, but its MD-MAE (3451) remains close to the best self-supervised values.

Restormer-Hybrid-RGS obtains comparable full-volume PSNR (22.98 dB) but weaker ROI PSNR (23.43 dB) than DRCNet at the fixed $K = 16$ operating point. Its FA-MAE (0.2238) is lower than DRCNet's (0.2575), while its MD-MAE (3440) is close to Patch2Self. Importantly, this comparison does not reflect Restormer's best-sweep configuration: the K sweep in Table 2 shows Restormer peaking at $K = 10$ with PSNR-ROI = 26.60 dB and FA-MAE = 0.2195, which is competitive with or better than DRCNet on both metrics. The gap at $K = 16$ is therefore partly a consequence of the fixed operating point rather than an inherent limitation of the Restormer backbone. DRCNet is also substantially faster at $K = 16$, requiring 34.1 s per target volume compared with 126.8 s for Restormer. The main comparison supports DRCNet-Hybrid-RGS as the more practical instantiation at this fixed operating point, while Restormer remains a useful higher-capacity probe for backbone-sensitivity analysis.

The supervised reference runs provide context for the cost of self-supervision and are not intended as rigorously tuned upper bounds. Both rows use the same DRCNet and Restormer architectures and training protocol as the self-supervised runs ($K = 16$, RGS gradient sampling, `seed` = 91021, `subset_fraction` = 0.6, Adam at 4.5×10^{-4}, best-loss checkpoint, single-subject D-Brain setting), with the sole difference that clean D-Brain phantom volumes are used as training targets instead of the noisy inputs. Supervised DRCNet reaches 22.92 dB PSNR-ROI, while supervised Restormer reaches 22.60 dB. In the present runs, these supervised reference models do not exceed the best self-supervised DRCNet result in image-domain metrics. This outcome should not be generalized as evidence that supervision is intrinsically inferior to self-supervision: the single-subject limited-data setting, Rician noise structure, and the absence of comparably tuned supervised hyperparameters all contribute. The more important observation is that the Hybrid RGS training strategy achieves competitive performance without requiring clean targets, which is essential for clinical and research DWI settings where paired noisy-clean acquisitions are unavailable or prohibitively expensive.
Note: †MD-MAE for D-Brain rows is reported in arbitrary units derived from the phantom intensity scale and should be interpreted as a relative comparison across methods, not as a physical diffusivity in mm^2/s.
TODO: Replace this planned qualitative comparison with final image panels

Table 7: Main quantitative comparison on D-Brain with Rician noise $\sigma = 0.1$. All Hybrid RGS methods use the fixed $K = 16$ cross-model operating point. ROI metrics are computed over brain tissue only (Ω_{ROI}). Time/vol. is inference time per target diffusion volume when available. Bold values indicate the best self-supervised neural or regression result in each metric column; italicized rows are supervised reference runs using the same architecture and training protocol but with clean phantom targets (not rigorously tuned upper bounds).

Method	PSNR ↑	SSIM ↑	MSE ↓	PSNR-ROI ↑	MSE-ROI ↓	FA-MAE ↓	MD-MAE† ↓	Time/vol. (s)
MP-PCA	22.59	0.4350	0.00551	28.44	0.00143	0.0898	4650	–
Patch2Self (DIPY OLS)	17.65	0.3806	0.01719	21.31	0.00739	0.2393	**3438**	–
MD-S2S (Conv2D)	15.14	0.4250	0.03061	15.28	0.02963	0.2298	3449	–
DRCNet-Hybrid-RGS	**23.90**	**0.4402**	**0.00408**	**26.93**	**0.00203**	0.2575	3451	**34.1**
Restormer-Hybrid-RGS	22.98	0.4288	0.00504	23.43	0.00454	**0.2238**	3440	126.8
DRCNet supervised	*21.37*	*0.3686*	*0.00729*	*22.92*	*0.00511*	*0.2387*	*3684*	*34.0*
Restormer supervised	*21.57*	*0.3696*	*0.00697*	*22.60*	*0.00549*	*0.2374*	*3713*	*125.3*

or remove the figure before submission. The final figure should include noisy input, MP-PCA, Patch2Self, MD-S2S, DRCNet-Hybrid-RGS, Restormer-Hybrid-RGS, reference image when available, and error maps.

5.2. Generalization to Real Scanner Noise

The D-Brain experiments provide controlled quantitative evaluation because synthetic Rician noise is added to data with an available reference. However, this setting cannot fully represent real scanner noise, which may include thermal noise, Gibbs ringing, eddy-current residuals, motion, coil-dependent spatial variation, and deviations from an ideal Rician model in low-SNR regions. We therefore evaluate Hybrid RGS on Stanford HARDI as a complementary real-data experiment. Stanford HARDI uses a different acquisition protocol from D-Brain, with $b = 2000$ s/mm^2 and $G = 150$ diffusion directions, compared with $b = 2500$ s/mm^2 and $G = 60$ in the primary synthetic experiments. Because only a single acquisition is available, no clean ground truth exists; this section therefore emphasizes qualitative assessment, downstream FA/MD map plausibility, and scalability rather than PSNR/SSIM against a reference.

The Stanford experiment trains the same self-supervised models directly on the noisy Stanford data. We use the same Hybrid RGS objective with $K = 16$, target-channel Bernoulli masking, and the same reconstruction loss; no clean targets are used. Patch2Self and MD-S2S are included as self-supervised baselines that are applicable without ground truth. MP-PCA can also be evaluated as a classical preprocessing method when available, but the central comparison is qualitative because all methods operate without an external reference. Evaluation focuses on visual inspection of denoised diffusion-weighted volumes, smoothness and anatomical plausibility of FA and MD maps, and reference-free stability measures such as variance within homogeneous tissue regions.

The Hybrid RGS models were successfully trained and applied to Stanford HARDI volumes. The method can be trained directly on real scanner data using the same self-supervised objective, and denoised outputs can be used to compute downstream tensor-derived maps. Visual assessment of denoised volumes and FA/MD maps requires the panels described in the TODO below; without those panels, Stanford results support feasibility and scalability rather than specific visual quality claims. In the current registry, FiLM-conditioned Stanford runs also compute tensor-summary errors against internal reference estimates, yielding FA-MAE values of 0.0356 for DRCNet-FiLM and 0.0756 for Restormer-FiLM; these numbers are not equivalent to ground-truth errors, but they indicate that the pipeline can produce stable diffusion-derived summaries on real scanner data.

TODO: Add visual Stanford FA/MD panels, residual maps, and/or homogeneous-ROI variance summaries. Without those panels, keep Stanford claims limited to feasibility and scalability.

Table 8 summarizes the Stanford K sweep using noisy-input PSNR-ROI, where the acquired noisy image is used as the comparison image. These values should not be compared with D-Brain PSNR and should not be interpreted as restoration accuracy; they measure closeness to the acquired signal. A method that changes the image very little can score well under this metric. For DRCNet, the noisy-input score increases from 34.86 dB at $K = 5$ to 39.47 dB at $K = 24$, then drops at $K = 30$, suggesting that very large input stacks are not necessarily beneficial. Restormer increases from 23.79 dB at $K = 5$ to 27.08 dB at $K = 30$, but remains more computationally expensive. Inference time scales moderately with K for DRCNet, from 23.5 to 28.0 s per volume, and remains around 84–85 s for Restormer. Thus, $K = 16$ remains a practical default for $G = 150$, while $K = 24$ may be explored when additional compute is available.

FiLM conditioning also transfers to Stanford without architectural changes. The FiLM-conditioned DRCNet run achieves a noisy-input PSNR-ROI of 36.55 dB with an inference time of 26.8 s per volume, while the FiLM-conditioned Restormer run achieves 24.00 dB with 85.4 s per volume. These values measure input closeness rather than denoising accuracy. Their main value is that target-direction metadata can be incorporated in a real, larger-shell protocol without breaking the self-supervised training pipeline.

Table 8: Stanford HARDI K sweep. PSNR-ROI is computed against the acquired noisy image and should be interpreted only as an input-closeness or signal-preservation measure rather than ground-truth restoration accuracy. Time/vol. is inference time per target diffusion volume.

Architecture	K	Noisy-input PSNR-ROI	Params (M)	Time/vol. (s)
DRCNet	5	34.86	0.106	23.5
DRCNet	10	36.99	0.111	25.3
DRCNet	16	38.07	0.116	25.2
DRCNet	24	39.47	0.123	26.3
DRCNet	30	35.18	0.128	28.0
Restormer	5	23.79	0.174	83.7
Restormer	10	26.60	0.176	84.4
Restormer	16	26.05	0.178	83.1
Restormer	24	25.65	0.180	84.7
Restormer	30	27.08	0.182	85.1

TODO: Replace the planned Stanford HARDI FA-map comparison with final image panels or remove the figure before submission. The final figure should show a representative axial slice crossing major white-matter tracts for noisy input, Patch2Self, MD-S2S, DRCNet-Hybrid-RGS, and Restormer-Hybrid-RGS.

Overall, the Stanford experiment supports the feasibility of Hybrid RGS beyond synthetic D-Brain corruption. The method can be trained directly on real scanner data, scales from $G = 60$ to $G = 150$ without architectural changes, and produces outputs suitable for qualitative FA/MD assessment. The main limitation is the absence of ground truth, which prevents claims of quantitative restoration accuracy. Future work should evaluate the same protocol on repeated acquisitions or test–retest datasets where one scan can serve as an approximate reference.

6. Discussion

The experiments support Hybrid RGS primarily as a self-supervised sampling-and-masking framework rather than as evidence for a single dominant architecture. Random gradient subsets improve over deterministic sequential windows, indicating that angular diversity acts as useful data augmentation during training. Moderate input sizes such as $K = 16$ provide a practical compromise between context and cost. FiLM conditioning is a promising but exploratory extension because deterministic acquisition metadata can be incorporated without weakening the blind-spot construction, but the current registry shows full-volume PSNR gains alongside ROI degradation. These findings identify the core contribution as the coupling of angular self-supervision with target-channel masking, which allows DW-MRI denoisers to be trained without clean targets and instantiated with different model families.

The results also clarify the role of architecture. The controlled capacity-and-dimensionality ablation at the fixed $K = 16$ operating point shows that lightweight 2D baselines (Plain-CNN-2D, Res-CNN-2D) outperform the current volumetric 3D models on most scalar image-domain metrics and on DRCNet tensor metrics, while Restormer3D retains a slight FA-MAE advantage. This comparison is specific to $K = 16$, which is DRCNet-favorable but not Restormer-optimal: the K sweep shows Restormer peaking at $K = 10$ with PSNR-ROI = 26.60 dB, exceeding Res-CNN-2D's 26.44 dB. The conclusion that the Hybrid RGS objective transfers effectively across backbones is robust, but claims about 2D baselines uniformly outperforming 3D volumetric variants are limited to the fixed $K = 16$ setting. Higher-capacity volumetric processing should be justified by additional evidence such as slice-to-slice coherence, tractography stability, or test–retest reproducibility rather than by the present scalar metrics at a single operating point.

MP-PCA remains a strong baseline, particularly for FA preservation under synthetic Rician noise. This is expected because local low-rank assumptions are well matched to spatially homogeneous simulated corruption. The proposed neural models improve substantially over Patch2Self and MD-S2S in image-domain ROI metrics, but they do not uniformly dominate every diffusion-derived metric. Reporting both image fidelity and tensor-derived errors is therefore essential: improvements in PSNR or MSE do not necessarily imply better FA, MD, AD, or RD estimates.

The framework has several limitations. First, the self-supervised objective relies on independence assumptions about noise across masked dimensions and diffusion volumes; correlated motion, Gibbs ringing, or reconstruction artifacts may violate these assumptions. Second, synthetic Rician experiments cannot fully represent real scanner noise, and Stanford HARDI lacks ground truth, so its results support feasibility rather than quantitative accuracy. Third, high-capacity networks and conditioning mechanisms may introduce anatomically plausible but quantitatively biased signal if validation is based only on image-domain similarity. Future work should therefore treat architecture choice as an empirical design variable and emphasize repeated acquisitions, test–retest datasets, tractography endpoints, and reader studies to determine when Hybrid RGS improves clinically relevant diffusion analysis.

7. Conclusion

This manuscript presented Hybrid RGS, an architecture-agnostic self-supervised framework for DW-MRI denoising based on random gradient subsets and voxel-wise target masking. The results indicate that the framework is effective as a training strategy: it improves over Patch2Self and MD-S2S in image-domain ROI metrics on D-Brain, benefits from random rather than sequential angular sampling, and remains stable across synthetic Rician noise levels. The experiments also show that MP-PCA is a highly competitive baseline for FA preservation under synthetic noise, emphasizing the need to evaluate both image-domain and diffusion-domain metrics. Architectural ablations suggest that the RGS objective transfers across DRCNet, Restormer-style, and lightweight 2D baselines (Plain-CNN-2D, Res-CNN-2D). At the fixed $K = 16$ cross-model operating point, the 2D baselines outperform the current volumetric 3D variants on most scalar metrics while Restormer3D retains a slight FA-MAE advantage; however, $K = 16$ is DRCNet-favorable and not Restormer-optimal—the K sweep shows Restormer peaking at $K = 10$ (PSNR-ROI = 26.60 dB), so the relative ranking between Res-CNN-2D and Restormer3D is operating-point dependent. Future work should therefore justify higher-capacity volumetric processing through qualitative coherence, tractography, or test–retest analyses, and should evaluate each backbone at its own optimal K. FiLM conditioning remains a lightweight but exploratory extension because it improves full-volume PSNR while degrading ROI PSNR in the current registry, and Stanford HARDI experiments demonstrate that Hybrid RGS can be trained directly on real scanner data with larger gradient shells and no clean reference.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Francisco Hernandez Sepulveda acknowledges support from a CONACyT (Consejo Nacional de Ciencia y Tecnología, Mexico) master's scholarship. The authors thank CIMAT (Centro de Investigacion en Matematicas), Guanajuato, Mexico, for providing institutional support and computational resources.

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